

1. Project Overview & Methodology

The objective of this project was to build a regression model which is capable of predicting house prices and identifying the underlying features that tune the real estate pricing.

How the Model Was Built:

The model pipeline was constructed by first cleaning the raw dataset, engineering new features (such as combining bedrooms and bathrooms into `total_rooms`), and one-hot encoding categorical text variables. To ensure the models didn't get heavily skewed by extreme luxury mansions, we trimmed the top and bottom 5% price outliers. Finally, we applied a logarithmic transformation to the target variable (price) to normalize its distribution, split the data into an 80/20 training/testing set, and evaluated both Linear Regression (Ridge) and a Random Forest Regressor.

2. Visual Insights & Data Exploration

Based on the generated visualizations, several key trends emerged:

- **Price Distribution:** After removing the extreme upper and lower 5% outliers and applying a log transformation, the house prices show a better normal distribution. This ensures the model is not giving too much weightage to outlier priced houses (too expensive too underpriced).
- **Feature Correlations :** The heatmap shows that the size of the property and premium amenities correlate highly with price spikes. The heatmap is essential for validating the engineered features.
- **Model Fit :** The scatter plot tracking Actual vs Predicted prices demonstrates a solid clustering around the diagonal reference line, indicating that our model's predictions scale appropriately across different price brackets without severe deviation.

Model Accuracy:

The Random Forest Regressor emerged as our most accurate model, achieving an **R^2 score of ~ 0.636** . So, the model can accurately explain about 63.6% of the reasons why house prices change in this dataset. On average, its predictions missed the actual price by about \$719,111 (MAE), which is high but still would give a decent performance given the natural variance in real estate pricing is sophisticated.